

Chapter 5

General Assumptions and Methodology

The objective of game-theoretic modeling is to sort precisely through the logic of a strategic situation: to explain behavior, and to predict or prescribe it.

We already know the language used to *describe* a game. This chapter covers the assumptions that let us go further and actually analyze one.

Modeling Trade-Offs

Writing down a game always means choosing how much realism to include. A simple model is easy to analyze; a complex one can capture more of what actually matters.

Suppose you wanted to model a couple negotiating how to split the evening's chores. At the initial node, one partner makes an offer; the other can say “okay” or counteroffer.

A coarse description might allow only “okay” or “I’ll do everything myself.” A finer one might track tone of voice, or a look that says more than words do.

Which level of detail belongs in the model depends on how much it matters strategically, weighed against how much harder it makes the game to solve.

Rationality

Definition. A player is **rational** if, whenever her action determines which of several outcomes occurs, she selects the action leading to the outcome she most prefers.

As in Chapter 4, we assume utility payoffs represent preferences over uncertain outcomes through their expectations. A rational player maximizes her own expected payoff, given her beliefs.

This is the assumption behind everything that follows. Once a game is on the table, we take it as given that every player is doing the best she can, given what she believes about the others.

Once beliefs are pinned down, rationality identifies the best choices, possibly with ties. The hard part of solving a game is working out which beliefs make sense, and that is what the rest of the course is about.

Rationality Does Not Mean Selfish

Rationality is a weak assumption: it accommodates any coherent set of preferences, not just a narrow desire for money.

“Coherent” mainly means *transitive*. If a player prefers A to B and B to C , she must prefer A to C ; otherwise no payoff numbers can represent her ranking at all.

An altruistic player, for example, has a payoff that increases in *other* players' money, not only her own. Maximizing that payoff is still perfectly rational; it just is not selfish.

Most examples in this course assume players care only about their own monetary payoff, since that keeps the mathematics simple. Watson's Chapter 25, which we skip, treats attitudes toward risk in full.

Example: Splitting a Red Envelope

At Lunar New Year, player 1 receives a NT\$1,000 red envelope and decides how much, $x \in \{0, 1, \dots, 1000\}$, to share with player 2, who can only accept. Economists call this one-sided game the **dictator game**.

1. If player 1 is entirely selfish, $u_1(x) = 1000 - x$, maximized at $x = 0$.
2. If she cares only about the worse-off person, $u_1(x) = \min\{x, 1000 - x\}$, maximized at $x = 500$: she equalizes the split.
3. If she cares only about player 2, $u_1(x) = x$, maximized at $x = 1000$: she gives everything away.

Every one of these players is rational. Each has a coherent ranking of outcomes and picks the action that leads to her top-ranked one; they simply want different things.

Discussion: The Lunch-Money Bet

Two students each hold NT\$100 of lunch money. Player 1 proposes matching pennies for it: each secretly picks Heads or Tails, and if the pennies match, player 1 takes player 2's NT\$100; if they differ, player 2 takes player 1's.

1. Write the normal form with payoffs as net gains in NT dollars.
2. Now suppose that, for each student, a fancy lunch (NT\$200) is worth $+1$ relative to an ordinary one (NT\$100), and going hungry (NT\$0) is worth -2 . Rewrite the matrix in these payoffs.
3. Player 2 believes player 1 will flip a fair coin to choose. Should player 2 agree to the bet at all?

Answer: The Lunch-Money Bet

Net gains (NT dollars):

	Heads	Tails
Heads	100, -100	-100, 100
Tails	-100, 100	100, -100

In utility:

	Heads	Tails
Heads	1, -2	-2, 1
Tails	-2, 1	1, -2

Against a coin flip, either action gives player 2 an expected utility of $-\frac{1}{2}$, since $\frac{1}{2}(1) + \frac{1}{2}(-2) = -\frac{1}{2}$. Declining gives 0, so she should decline.

In dollars, the bet has an expected value of exactly 0, so a player whose payoff *is* money would be indifferent. The utility payoffs encode risk aversion: utility is a concave function of money, so losing hurts more than winning helps. Rationality is compatible with both attitudes; what matters is that the payoffs in the matrix are the ones the player actually maximizes.

Common Knowledge

Every solution concept in this course assumes the players share an understanding of the game itself.

Definition. A fact F is **common knowledge** between the players if each player knows F , each player knows that the others know F , each player knows that the others know that each player knows F , and so on, without end.

It is as if the players sat around a table on which the entire game was laid out, so everyone can verify that everyone else sees it too.

Common knowledge of the game does not mean common knowledge of where the players are *within* it: a non-singleton information set means the player moving there cannot distinguish its nodes.

Why the Game Must Be Common Knowledge

Suppose player 2 is not sure whether she is playing Matching Pennies or the Prisoners' Dilemma. That uncertainty is part of her strategic situation, so it belongs in the description: the true game is a different, larger one.

The same holds if player 1 is not sure that player 2 knows player 1's payoffs. Part IV of the course builds exactly such games, with random events and incomplete information written into the tree.

Until then, we take the game as common knowledge. Experimenters achieve this by reading the instructions aloud with every subject in the same room, which is the “game laid out on the table” made literal.

Discussion: Common Knowledge at Disney

Recall the Katzenberg and Eisner game from Chapter 2. Katzenberg leaves Disney, Eisner decides whether to produce *A Bug's Life*, and Katzenberg then decides whether to produce *Antz* without observing Eisner's choice.

At the moment Katzenberg decides on *Antz*, which of these facts are common knowledge between the two players?

1. The tree, including all payoffs.
2. That Katzenberg has left Disney.
3. Whether Eisner is producing *A Bug's Life*.

Answer: Common Knowledge at Disney

1. Yes, by assumption. Every analysis in this course starts from the game itself being common knowledge.
2. Yes. Eisner's decision node is a singleton: he observed Katzenberg leave, Katzenberg knows Eisner observed it, Eisner knows that Katzenberg knows, and so on. Nothing in the tree breaks the chain.
3. No. Eisner knows his own choice, but Katzenberg's two nodes sit in one information set: he does not know which one he is at. The fact is known to one player and not the other, so it is not even mutual knowledge, let alone common.

This is the distinction from the previous slide in a concrete game: common knowledge of the *game* is assumed throughout; Katzenberg does not know Eisner's action at this information set.

Discussion: The LINE Message

Jack messages Jill on LINE that restaurant A is closed, and suggests restaurant B instead. Jill reads it and replies, “Got it, see you at B.” Assume messages can be delayed and neither person has a reliable read receipt.

Jill arrives at B. Jack is not there.

Is Jack just running late? Or did he never see Jill’s reply, and is he still waiting somewhere else, worried that *she* might not have gotten *his* message?

Which levels of knowledge have been established? How would a reliably public announcement change the situation?

Answer: The LINE Message

Both know that A is closed: Jack knew it already, and Jill learned it from his message. Jill also knows that Jack knows.

Until Jack reads Jill's reply, he does not know that she received his message. Even after he reads it, Jill does not know that he has read her reply. Further acknowledgments can add levels of knowledge, but their delivery is still uncertain.

Thus, knowing that A is closed is different from having common knowledge of it, or of an agreement to meet at B.

An ideal public announcement is heard by both, and its reliable reception is itself common knowledge. A phone call can approximate this; real-time speech alone does not guarantee it.

Discussion: The Basketball Buzzer-Beater

Consider a simplified review of a buzzer-beater at a campus basketball game. One official checks that the ball left the shooter's hand before time expired; another checks that the shooter was in bounds.

Under this review procedure, either failed check rejects the basket, while both checks must pass to confirm it.

Why might confirming the basket require more communication than rejecting it?

Suppose the final decision is announced publicly. Under what assumptions would it become common knowledge?

Answer: The Basketball Buzzer-Beater

One failed condition is enough to reject the basket. An official who observes a failure can report it immediately.

Confirmation requires combining the two officials' information: the shot was on time *and* the shooter was in bounds. Neither official's own check establishes both facts.

Sharing information settles whether both conditions hold. Common knowledge asks more: does everyone know the decision, know that everyone else knows it, and so on?

A final announcement makes the decision common knowledge if everyone hears it and its reliable, public reception is itself common knowledge.

Discussion: The Hats

Ten people stand in a circle wearing red or black hats. Each sees every other hat but not her own. A host publicly announces, truthfully, that at least one hat is black.

The host then calls rounds 1, 2, and so on. At each call, everyone simultaneously shouts “Black!” if she can deduce her hat is black from what she saw and heard *before* that round; otherwise she stays silent.

These rules, perfect reasoning, and everyone’s ability to hear all responses are common knowledge. No one guesses or gives other clues.

If exactly n hats are black, why do their wearers first shout in round n ? For $n \geq 2$, what did the announcement add?

Answer: The Hats

If $n = 1$: the one black-hatted person sees nine red hats. Since the host said at least one hat is black, she knows it must be hers, and shouts in round 1.

If $n = 2$: each black-hatted person sees exactly one black hat. Were that the *only* one, its wearer would have shouted in round 1. When no one does, each learns her own hat is black too, and both shout in round 2.

The pattern continues by induction: with n black hats, each wearer sees $n - 1$ others and, after their silence through round $n - 1$, shouts in round n . Red-hatted people see n black hats and cannot yet conclude their own hat is black.

For $n \geq 2$, everyone could already see a black hat, so the announcement adds no new first-order fact about the hats. What it adds is that “at least one hat is black” is now *common* knowledge, which starts the chain.

Common Knowledge of Rationality

The hats puzzle needed common knowledge of the reasoning and response rules. Each person relied on others drawing the right conclusions, and on them relying on her, and so on.

In a strategic game, the corresponding assumption is common knowledge of *rationality*: everyone maximizes expected payoff given her beliefs, everyone knows this, and so on. Perfect logical reasoning in the puzzle illustrates this hierarchy; it is not itself the definition of rationality.

Together with common knowledge of the game, this leads to rationalizability, our first solution concept. It need not select a single outcome.

Overview of Solution Concepts

Studying rational behavior splits into two tasks: defining precisely what rationality requires in a given setting, and applying that definition to specific games. The theories that result are called **solution concepts**.

The course develops these concepts, then refines or extends them to richer settings.

1. Rationalizability builds on common knowledge of the game and rationality; we compute it by iterated strict dominance in two-player games.
2. Nash equilibrium adds that beliefs and actual behavior line up.
3. Subgame-perfect equilibrium refines this for sequential settings.
4. Bayesian Nash and perfect Bayesian equilibrium extend it to games with random events and incomplete information.

The Issue of Realism

Full rationality can demand a great deal: sophisticated beliefs, and the calculation needed to best-respond to them.

Real behavior sometimes falls short, systematically. Players may not share an understanding of the game, may be cognitively biased, or may be unable to carry out the required calculations at all. And a perfectly rational player may still doubt that the *others* are.

A model that departs from reality still earns its keep two ways: normatively, by showing people how to play better, and descriptively, once enriched with bounded rationality, by explaining how people actually do play.

Example: Bounded Rationality

The numbers 1 through 9 sit in the center of a table. Players alternate taking one number for their own side. The first player whose collection contains three numbers summing to 15 wins. If all numbers are taken without a winner, the game is a draw.

2	7	6
9	5	1
4	3	8

This is exactly tic-tac-toe in disguise: the nine numbers form a magic square, so “three numbers summing to 15” is the same as “three numbers in a line.” Optimal play on both sides leads to a draw. Recognizing the disguise makes the required reasoning much easier.

Chapter in a Nutshell

- Building a game means trading off realism against tractability; there is no single right level of detail.
- Rationality means maximizing expected payoff given one's preferences, which must be transitive but need not be selfish or risk-neutral.
- Common knowledge, not mere knowledge, is what conventional analysis of games assumes and what the hats and LINE-message puzzles turn on.
- Solution concepts refine or extend our assumptions about behavior and beliefs; models built on full rationality are still useful even where real behavior falls short of it.

Key Concepts

- Rationality is choosing the most preferred outcome, formalized as maximizing expected payoff.
- Common knowledge means everyone knows a fact, knows that everyone knows it, and so on without end.
- Solution concept is the general term for a theory of how a game is, or should be, played.
- Bounded rationality describes systematic ways real behavior departs from full rationality.